

# MoCHII: Physics, Atomic Data, Algorithms, and Validation

## Abstract

MoCHII (MONte Carlo for H II regions) adds a photoionization gas layer to the validated octree-AMR dust radiative-transfer engine of MoCafe v2.00. This memo documents everything built through Stage G3: the ionizing frequency band and its zero-variance transport, the complete atomic-data inventory with sources and the analytic fitting functions evaluated at run time, the iteration and equilibrium algorithms (ionization, thermal balance, trace-metal cascade, explicit diffuse field), the Tier-2  $n$ -level emissivity solver, the ported nebular continuum and Storey & Hummer recombination lines, and the quantitative validation gates: G0 (rate integrals vs. analytic attenuation, bias +0.004%), G1 (Strömgren sphere vs. an exact 1D reference, +0.56% in ionized volume; AMR  $\equiv$  uniform), G2a (thermal H/He nebula, median 0.14% in  $T_e$ ), G2c (MOCASSIN Lexington HII20  $T_e$ (H II) +0.11%; HII40 structure validated with the KF96 charge-exchange addition,  $T_e$  +7% attributed to atomic-data vintage), G3 (case A/B bracketing), and the line-flux comparison ( $H\beta$  absolute  $-0.7\%$  at HII40). Also documented: the G4 solution-driven I-front re-refinement (re-refined vs. native high-resolution octree +0.008% in ionized volume with  $4.5\times$  fewer cells) and an Illustris-TNG post-processing demonstration; the completed dust coupling (ionizing-band dust absorption with the dusty Strömgren reduction 0.735 reproduced to +0.69%; EUV/FUV dust *scattering* with forced-first interaction sampling, validated by bracketing; live ionization-dependent dust with a PAH-specific survival factor; and the equilibrium dust temperature + IR spectrum consuming the transported grain heating, energy closure 1.000); and the G5 argon addition as a pure data operation.

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## 1. Scope and design rules

MoCHII copies validated modules from MoCafe v2.00 (octree, event-driven traversal, MPI shared memory, HDF5/FITS I/O) and adds gas microphysics under three rules fixed in docs/PLAN.md: (1) *atomic data are fitted offline, evaluated online* — the Python pipeline under tools/fitting/ is the auditable source of every coefficient in data/atomic/, and Fortran only evaluates closed forms; (2) *adding an ion is a data operation*, realized by the species registry (Section 4.4); (3) *validation by staged gates*, each stage quantitatively compared against an analytic or external reference before the next begins.

The stages covered here: **G0** ionizing band + rate integrals (no feedback); **G1** ionization equilibrium + opacity feedback (case B, fixed  $T_e$ ); **G2** thermal balance + trace-metal frame; **G3** explicit diffuse field (case A); **G4** solution-driven I-front

re-refinement + snapshot post-processing; the **dust-coupling** stages (PLAN section 7: absorption competition, scattering, live PAH split, grain heating/temperature, and the SEDust stochastic-emission coupling); **G5** incremental ions (argon, magnesium, iron); plus the output-time diagnostic layer: Tier-2  $n$ -level emissivities, Storey & Hummer recombination lines, and the nebular continuum.

## 2. The ionizing band and transport

### 2.1 Frequency bins and the source

The ionizing band covers photon energies  $[E_{\min}, E_{\max}] = [13.598, 100]$  eV with  $n_\nu$  log-spaced bins (`par%nnu_ion`; gates use 32). Bin centers are geometric means; the source spectrum (Planck  $B_\nu(T_\star)$  or a 2-column file) is integrated on a 32-point sub-grid within each bin and normalized so that  $\sum_b L_b = L_{\text{band}}$  (`par%luminosity`). Packets sample the bin from the luminosity CDF and carry  $L_{\text{pkt}} = L_{\text{band}}/N$ . Bin-count convergence of the rate integrals (measured in the G0 gate): 16 bins  $\rightarrow$  2.2%, 32  $\rightarrow$  2.1%, 64  $\rightarrow$  0.02% against a 2048-bin reference.

**FUV extension.** With `par%add_fuv` the band grows a second, lower segment: `par%nnu_fuv` log bins on  $[E_{\text{FUV}}, E_{\min}]$  (`par%efuv_min`, default 6 eV) *below* the unchanged ionizing bins, so widening the band does not coarsen the ionizing resolution (the earlier single-segment caveat). `par%luminosity` keeps its meaning as the ionizing  $[E_{\min}, E_{\max}]$  luminosity; the FUV part of the same source spectrum rides on top ( $L_{\text{FUV}}/L_{\text{ion}} = 1.097$  for a  $4 \times 10^4$  K Planck source at 6 eV), so the ionizing photon budget  $Q_{\text{H}}$  is preserved without manual band-ratio rescaling and packets carry  $L_{\text{pkt}} = (L_{\text{ion}} + L_{\text{FUV}})/N$ . H and He cross sections vanish below their thresholds, so the FUV bins interact only with dust and with the low-threshold metal stages (next subsection) — they penetrate the neutral gas beyond the I-front and photoionize Mg I (7.65 eV), C I (11.26 eV), S I (10.36 eV), and Fe I (7.90 eV) there, producing the PDR-side Mg II and C II that the ionizing-only band cannot.

### 2.2 Opacity and the zero-variance edge walk

Each leaf carries the band absorption coefficient per code length

$$\kappa_b = n_{\text{H}} [x_{\text{HI}} \sigma_{\text{HI}}(E_b) + y_{\text{He}} (x_{\text{HeI}} \sigma_{\text{HeI}} + x_{\text{HeII}} \sigma_{\text{HeII}})(E_b)] d_{\text{cm}} (+ \kappa_{\text{dust}}), \quad (1)$$

stored as the full  $(n_\nu, n_{\text{leaf}})$  array `kap_ion` — the grey  $s_{\text{ext}}$  rescaling of the dust SED band cannot apply because the gas opacity varies with both leaf and bin. With `par%ion_metal_abs` (default on when `use_metals`) the registry metals join the absorption,  $\Delta\kappa_b = n_{\text{H}} \sum_{\text{el}} A_{\text{el}} \sum_i f_i \sigma_{\text{VFKY96},i}(E_b) d_{\text{cm}}$ , with the stage fractions  $f_i$  from the same product chain as the cooling, iterated by the opacity feedback exactly like  $x_{\text{HI}}$ . Above 13.6 eV this is a  $\sim 10^{-22}$  cm<sup>2</sup> per H perturbation next to H/He and dust; in the FUV bins it is the only *gas* opacity. The absorbed share returns to the metals through their own  $\Gamma$  integrals, so the bookkeeping stays consistent (metal photoheating is not added to the thermal balance — negligible at trace abundances).

Through G3 the band has no scattering and the diffuse field is emitted explicitly, so every packet's contribution to the

mean-intensity tally is computed *analytically* along its ray (the forced-first-flight estimator of MoCafe, made exact):

$$\Delta J_b(\text{leaf}) = w L_{\text{pkt}} \frac{e^{-\tau_{\text{in}}} - e^{-\tau_{\text{out}}}}{\kappa_b}, \quad (2)$$

accumulated leaf by leaf until the domain edge (`raytrace_ion_to_edge_amr`). No interaction sampling is needed; the only Monte Carlo variance is in direction and bin sampling. The G0 gate verifies this estimator to a mean bias of +0.004%.

### 2.3 Rate integrals

With  $V$  the leaf volume and  $J_b$  the reduced tally, the code forms for each absorber  $i$

$$\Gamma_i = \sum_b \frac{[4\pi J \Delta\nu]_b \sigma_i(E_b)}{h\nu_b}, \quad \mathcal{H}_i = \sum_b [4\pi J \Delta\nu]_b \sigma_i(E_b) \left(1 - \frac{E_{\text{th},i}}{E_b}\right), \quad (3)$$

the photoionization rate [ $\text{s}^{-1}$ ] and photoheating [ $\text{erg s}^{-1}$ ] for each particle of species  $i$ .

## 3. Atomic data

Every coefficient file in `data/atomic/` carries a provenance header (source, version, fit range, maximum fit error). The generators live in `tools/fitting/`.

### 3.1 Photoionization cross sections (Verner et al. 1996)

All cross sections use the full VF KY96 single-shell analytic form (Verner, Ferland, Korista & Yakovlev 1996, ApJ 465, 487, Eq. 1):

$$x = \frac{E}{E_0} - y_0, \quad z = \sqrt{x^2 + y_1^2}, \quad Q = 5.5 - \frac{1}{2}P, \quad \sigma(E) = \sigma_0 [(x-1)^2 + y_w^2] z^{-Q} \left(1 + \sqrt{z/y_a}\right)^{-P}, \quad (4)$$

zero below threshold,  $\sigma_0$  in Mb. Parameters were parsed directly from the authors' `phfit2.f` PH2 (outer-shell) table and verified against the EXHALE implementation; H/He values are compiled in `photo_xsec.f90`, the registry ions in `element_<el>.txt`. For the elements *absent* from the VF KY96 outer-shell table (P, Cl, K — Opacity-Project gaps; the MOCASSIN `ph2.dat` read loop skips the same  $Z$ ), the Verner & Yakovlev (1995) outer-shell subshell fit is used instead (`sigma_vy95`; PHOT02 lines):  $\sigma = \sigma_0[(y-1)^2 + y_w^2] y^{-(5.5+l-P/2)} (1 + \sqrt{y/y_a})^{-P}$  with  $y = E/E_0$  and  $l$  the subshell orbital quantum number — the same VF KY96/VY95 hybrid as MOCASSIN `phfitEL`. Chlorine is the first user of this path:

| ion   | $E_{\text{th}}[\text{eV}]$ | $E_0$  | $\sigma_0[\text{Mb}]$ | $y_a$ | $P$   | $y_w$ | $y_0$  | $y_1$ |
|-------|----------------------------|--------|-----------------------|-------|-------|-------|--------|-------|
| H I   | 13.598                     | 0.4298 | $5.475 \times 10^4$   | 32.88 | 2.963 | 0     | 0      | 0     |
| He I  | 24.587                     | 13.61  | 949.2                 | 1.469 | 3.188 | 2.039 | 0.4434 | 2.136 |
| He II | 54.416                     | 1.720  | $1.369 \times 10^4$   | 32.88 | 2.963 | 0     | 0      | 0     |

plus C I–III, N I–II, O I–III, Ne I–III, S I–IV, Ar I–IV (20 rows, in the element files). The G0 gate cross-checks the curves against EXHALE's independent implementations (the H I fit agrees with the exact nonrelativistic hydrogenic formula to 0.72%; thresholds  $\sigma_{\text{HI}} = 6.35$ ,  $\sigma_{\text{HeI}} = 7.43$ ,  $\sigma_{\text{HeII}} = 1.59$  Mb).

### 3.2 H/He recombination (Hui & Gnedin 1997)

Case A/B radiative recombination uses the Hui & Gnedin (1997, MNRAS 292, 27) fits with  $\lambda = 2T_{\text{TR}}/T$  ( $T_{\text{TR}} = 157807$ , 285335, 631515 K for H I, He I, He II):

$$\begin{aligned}\alpha_A^{\text{HII}} &= 1.269 \times 10^{-13} \lambda^{1.503} / [1 + (\lambda/0.522)^{0.470}]^{1.923}, & \alpha_B^{\text{HII}} &= 2.753 \times 10^{-14} \lambda^{1.500} / [1 + (\lambda/2.740)^{0.407}]^{2.242}, \\ \alpha_A^{\text{HeII}} &= 3.000 \times 10^{-14} \lambda^{0.654}, & \alpha_B^{\text{HeII}} &= 1.260 \times 10^{-14} \lambda^{0.750},\end{aligned}\quad (5)$$

and He III hydrogenic ( $Z=2$ ): the H fit on  $\lambda(T_{\text{TR}}^{\text{HeII}})$  times  $Z$ . At  $10^4$  K,  $\alpha_B^{\text{HII}} = 2.594 \times 10^{-13} \text{ cm}^3 \text{ s}^{-1}$  (canonical). Recombination *cooling* uses the companion fits

$$\beta_B^{\text{HII}} = 3.435 \times 10^{-30} T \lambda^{1.970} / [1 + (\lambda/2.250)^{0.376}]^{3.720} \text{ erg cm}^3 \text{ s}^{-1}, \quad (6)$$

(case A analog with  $1.778 \times 10^{-29}$ , 0.541, 1.965, 0.502, 2.697), He III scaled as for  $\alpha$ , and He II as  $k_B T \alpha_{A,B}^{\text{HeII}}$  (their recommendation; He II dielectronic recombination is negligible at nebular temperatures).

### 3.3 Collisional ionization (Voronov 1997)

$$k(T) = A \frac{1 + P\sqrt{U}}{X + U} U^K e^{-U}, \quad U = \frac{\Delta E}{kT}, \quad (7)$$

with parameters ( $\Delta E$  [eV],  $P, A, X, K$ ) parsed from the author's `cf.it.f` table:

| ion   | $\Delta E$ | $P$ | $A [\text{cm}^3 \text{ s}^{-1}]$ | $X$   | $K$  |
|-------|------------|-----|----------------------------------|-------|------|
| H I   | 13.6       | 0   | $2.91 \times 10^{-8}$            | 0.232 | 0.39 |
| He I  | 24.6       | 0   | $1.75 \times 10^{-8}$            | 0.180 | 0.35 |
| He II | 54.4       | 1   | $2.05 \times 10^{-9}$            | 0.265 | 0.25 |

plus the 16 registry rows in the element files.

### 3.4 Metal recombination: Badnell RR + DR

For the registry cascade, total recombination of the recombining ion is Badnell (2006, ApJS 167, 334) radiative recombination

$$\alpha_{\text{RR}}(T) = A \left[ \sqrt{T/T_0} (1 + \sqrt{T/T_0})^{1-b} (1 + \sqrt{T/T_1})^{1+b} \right]^{-1}, \quad b = B + C e^{-T_2/T}, \quad (8)$$

plus the Badnell-group dielectronic fits  $\alpha_{\text{DR}}(T) = T^{-3/2} \sum_i c_i e^{-E_i/T}$ . Coefficients are parsed from the CHIANTI v11.0.2 `.rrparams/` `.drparams` files of each recombining ion; the parsed values were verified *identical* to the independently transcribed Badnell table in EXHALE (e.g. O II:  $A = 6.622 \times 10^{-11}$ ,  $B = 0.6109$ ,  $T_0 = 4.136$ ,  $T_1 = 4.214 \times 10^6$ ,  $C = 0.4093$ ,  $T_2 = 8.770 \times 10^4$ ). Where CHIANTI carries no Badnell DR (Ar II, Ar III), the `.drparams` type-2 entries are the Shull & Van Steenberg (1982) form  $\alpha_{\text{DR}} = A T^{-3/2} e^{-T_0/T} (1 + B e^{-T_1/T})$ , written as DR2 lines and supported by the same reader (negligible at nebular temperatures; kept for completeness). One further CHIANTI RR type appears with iron: the Fe II `.rrparams` is a type-3 power law, written as RR2 lines and evaluated as  $\alpha_{\text{RR}} = A (T/10^4 \text{ K})^{-\eta}$ .

### 3.5 Charge exchange with hydrogen

First-stage reactions use the verified transcription of Huang et al. (2023, ApJ 951, 123) Table 4 (sources Stancil et al. 1998/1999, Lin et al. 2005, Zhao et al. 2005, Kingdon & Ferland 1996), e.g. for oxygen

$$\begin{aligned} k(\text{O} + \text{H}^+) &= 2.08 \times 10^{-9} T_4^{0.405} + 1.11 \times 10^{-11} T_4^{-0.458}, \\ k(\text{O}^+ + \text{H}) &= (1.26 \times 10^{-9} T_4^{0.517} + 4.25 \times 10^{-10} T_4^{0.00669}) e^{-227/T}. \end{aligned} \quad (9)$$

Higher stages ( $\text{X}^{2+}$ ,  $\text{X}^{3+} + \text{H}^0$ , recombination direction) use the Kingdon & Ferland (1996) fits  $k = a T_4^b (1 + c e^{dT_4})$ , transcribed from the MOCASSIN `update_mod.f90` chex table and evaluated at  $T$  clamped into the fit validity ( $\sim 5\,000$ – $50\,000$  K):

| reaction                    | $a$ [ $10^{-9} \text{cm}^3 \text{s}^{-1}$ ] | $b$                   | $c$    | $d$                    |
|-----------------------------|---------------------------------------------|-----------------------|--------|------------------------|
| $\text{C}^{2+} + \text{H}$  | 3.25                                        | 0.21                  | 0.19   | −3.29                  |
| $\text{C}^{3+} + \text{H}$  | 332.46                                      | −0.11                 | −0.995 | $−1.58 \times 10^{-3}$ |
| $\text{N}^{2+} + \text{H}$  | 4.54                                        | 0.57                  | −0.65  | −0.89                  |
| $\text{O}^{2+} + \text{H}$  | 3.98                                        | 0.26                  | 0.56   | −2.62                  |
| $\text{Ne}^{2+} + \text{H}$ | 14.73                                       | 0.0452                | −0.84  | −0.31                  |
| $\text{S}^{2+} + \text{H}$  | 2.29                                        | 0.0402                | 1.59   | −6.06                  |
| $\text{S}^{3+} + \text{H}$  | 6.44                                        | 0.13                  | 2.69   | −5.69                  |
| $\text{Ar}^{2+} + \text{H}$ | 4.57                                        | 0.27                  | −0.18  | −1.57                  |
| $\text{Ar}^{3+} + \text{H}$ | 6.37                                        | 2.12                  | 10.21  | −6.22                  |
| $\text{Mg}^{2+} + \text{H}$ | $8.58 \times 10^{-5}$                       | $2.49 \times 10^{-3}$ | 0.0293 | −4.33                  |
| $\text{Fe}^{2+} + \text{H}$ | 1.26                                        | 0.0772                | −0.41  | −7.31                  |
| $\text{Fe}^{3+} + \text{H}$ | 3.42                                        | 0.51                  | −2.06  | −8.99                  |

(A transcription slip in the MOCASSIN chex table assigns its  $\text{Ar}^{2+}$  slot twice, the second time with the  $\text{Ca}^{2+}$  values; the rows above are the KF96 argon entries.) The Mg and Fe first-stage reactions follow the Huang et al. Table 4 transcription:  $\text{Mg} + \text{H}^+$  is the KF96 form  $k = a T_4^b (1 + c e^{dT_4})$  and the reverse direction carries an added barrier factor  $e^{-6.91/T_4}$  (i.e.  $6.91 \times 10^4$  K, endothermic);  $\text{Fe} + \text{H}^+$  is a fast constant  $4.0 \times 10^{-9} \text{cm}^3 \text{s}^{-1}$  and the reverse is  $a(T/300)^b e^{-6.61/T_4}$ . A stage-attribution caveat: for Mg and Fe the MOCASSIN chex transcription places these same coefficient rows one ionization stage higher than the Huang Table 4 labels; the Huang labels are adopted (documented in `make_element_data.py`). These proved decisive for the HII40 gate: in the O III zone  $n_{\text{HI}} \simeq 0.5 \text{cm}^{-3}$  makes  $k_{\text{CX}} n_{\text{HI}}$  comparable to  $\alpha n_e$  (Section 5.4).

### 3.6 Tier-1 cooling fits

Collisional line cooling for the thermal loop uses the EXHALE multi-exponential form fitted to direct CHIANTI v11.0.2 evaluations (`fit_cooling_tier1.py`; reader `chianti_cooling.py` parses `.elvlc/.scups` with Burgess & Tully 1992 descaling):

$$\Lambda_{\text{ion}}(T) = T^{-1/2} \sum_{i=1}^N A_i e^{-T_i/T} \quad [\text{erg cm}^3 \text{s}^{-1} \text{ per } (n_e n_{\text{ion}})], \quad (10)$$

lower levels Boltzmann-distributed over the ground-term fine structure (within 0.4 eV), optically thin, low-density limit. Fit range  $10^3$ – $10^5$  K; Levenberg–Marquardt on  $\log \Lambda$  with the  $T_i$  ladder initialized from the level energies.

| ion   | $N$ | max err            | ion    | $N$ | max err            |
|-------|-----|--------------------|--------|-----|--------------------|
| H I   | 4   | 0.52%              | O I    | 6   | 0.22%              |
| C I   | 6   | 0.34%              | O II   | 5   | 0.07%              |
| C II  | 6   | 0.11%              | O III  | 4   | 1.83%              |
| C III | 5   | 0.58% <sup>a</sup> | S II   | 3   | 2.26%              |
| N I   | 5   | 0.59%              | S III  | 6   | 0.46%              |
| N II  | 6   | 0.14%              | Ne II  | 7   | 0.65%              |
| N III | 6   | 0.15%              | Ne III | 4   | 0.58%              |
|       |     |                    | Ar III | 5   | 1.21% <sup>b</sup> |
|       |     |                    | Ar IV  | 5   | 0.25%              |

<sup>a</sup>in the relevant range ( $\Lambda > 10^{-3}$  max); 5.4% at the negligible low- $T$  tail. <sup>b</sup>over 5 000–20 000 K; 4.6% at the  $10^5$ -K endpoint.

Ne II is a single fine-structure line (12.8  $\mu\text{m}$ ) whose  $\sim T^{-0.35}$  decay requires a ladder of  $T_i$  terms to superpose. The caveat carried in every header: these are low-density-limit fits; density-dependent emissivities come from Tier 2.

### 3.7 Tier-2 $n$ -level data

For output-time diagnostics, `fit_nlevel_tier2.py` exports for each ion the lowest  $n$  levels (energies, statistical weights), the radiative  $A$ -values (`.wgfa`), and the effective collision strengths  $\Upsilon(T)$  as low-order Chebyshev fits in Burgess–Tully scaled space: with  $e_i = kT/\Delta E$ ,

$$s = \begin{cases} 1 - \ln C / \ln(e_i + C) & \text{types 1, 4} \\ e_i / (e_i + C) & \text{types 2, 3, 5, 6} \end{cases}, \quad u = \frac{2(s - s_{\text{lo}})}{s_{\text{hi}} - s_{\text{lo}}} - 1, \quad y = \sum_k c_k T_k(u), \quad (11)$$

descaled by type ( $y \ln(e_i + e)$ ;  $y$ ;  $y/(e_i + 1)$ ;  $y \ln(e_i + C)$ ;  $y/e_i$ ;  $10^y$ ), with a log flag storing  $\log_{10} y$  for weak peaked intercombination transitions. Thirteen ions are exported (O I–III, N II–III, C II–III, S II–III, Ne II–III, Ar III–IV); the worst transition fit error is 1.7% (O III  $^1\text{S}_0$ – $^5\text{S}_2$ ,  $\Upsilon \sim 10^{-4}$ ), typically  $\lesssim 0.5\%$ . Ar II has no level/collision data in CHIANTI v11 (only RR/DR), so that stage is tracked without cooling or lines ([Ar II] 6.98  $\mu\text{m}$  omitted). Verification is layered: the fitted files reproduce direct CHIANTI spline evaluation of the classic ratios to 0.05–0.5%; the PyNeb cross-check gives [O III] 5007/4363 within 1.2% and [N II] 6584/5755 within 0.3% (the density diagnostics differ by 6–10% at  $10^4 \text{ cm}^{-3}$  from PyNeb’s independent collision data — database spread); and the Fortran solver (Section 4.12) matches the Python reference implementation to four significant digits on 26 lines.

### 3.8 Free-free Gaunt factors and continuum data

`gaunt.f90` is the verbatim MOCASSIN copy of the Hummer (1988) two-dimensional Chebyshev fit of the Karzas & Latter (1961) free-free Gaunt factor ( $\leq 0.7\%$ ; interface `getGauntFF(z, log10Te, xlf, g)`). The thermal loop uses the thermally averaged  $\bar{g}(T, Z) = \int g_{\text{ff}}(u) e^{-u} du$  tabulated at setup ( $\bar{g}(10^4 \text{ K}, Z=1) = 1.2626$ ). The van Hoof et al. (2014) table is now available as a drop-in (`par%gaunt_vh14; data/gauntff_vh14.dat`, an  $81 \times 146$  grid in  $\log_{10} \gamma^2 = \log_{10}(Z^2 \text{ Ry}/kT)$  and  $\log_{10} u = \log_{10}(h\nu/kT)$ , selected at both call sites — the cooling average and the continuum  $\gamma_{\text{ff}}$ ). The two compilations

agree: switching moves  $T_e$  by a median  $5 \times 10^{-5}$ , at most 0.5%, consistent with the sub-percent Hummer accuracy van Hoof et al. report. Hummer remains the default so the recorded gates reproduce.

The nebular continuum tables `data/gamma{HI,HeI,HeII}.dat` (MOCASSIN inheritance; 21 temperatures  $\times$  paired-edge frequency points) were *numerically verified* (2026-07-12) against a hydrogenic Kramers Milne sum: they contain free-bound *plus* free-free combined coefficients (at  $\nu = 0.0069$  Ryd,  $10^4$  K the table value 9.47 decomposes into  $\text{ff} \simeq 8.9 + \text{fb} \simeq 0.6$  in  $10^{-40}$  units; near the Balmer edge the residual is  $\sim 10\%$ , the size of the neglected bound-free Gaunt factor). Consequently the port never adds free-free inside the table range. The Ercolano & Storey (2006) item is resolved as *provenance*: the MOCASSIN 2.02 tables ship in the ES06 compact paired-edge format, so no swap is needed — the verification above is the record. Two-photon continua follow MOCASSIN: Nussbaumer & Schmutz (1984) spectral shape with Pengelly (1964)  $\alpha_{\text{eff}}(2s)$  and the Osterbrock  $q(2^2\text{S}-2^2\text{P})$  collisional suppression for H I; the Z-scaled form with Storey & Hummer  $\alpha_{\text{eff}}$  for He II; Almog & Netzer (1989) shape (`data/HeI2phot.dat`) with Benjamin, Skillman & Smits (1999)  $\alpha_{\text{eff}}$  for He I.

### 3.9 H I and He II recombination lines (Storey & Hummer 1995)

`make_sh95_lines.py` parses the SH95 case-B data (`e1bx.d`) with the reference reader indexing  $k = (2n_{\text{top}} - n_{\text{ll}} - n_{\text{l}} + 1)(n_{\text{l}} - n_{\text{ll}})/2 + n_{\text{top}} - n_{\text{u}} + 1$  and exports  $\text{H}\alpha$ ,  $\text{H}\beta$ ,  $\text{H}\gamma$ ,  $\text{H}\delta$ ,  $\text{P}\alpha$ ,  $\text{P}\beta$ ,  $\text{Br}\gamma$  on the full  $(T, n_e)$  grid ( $11 \times 13$ ) to `data/atomic/sh95_hi_caseB.txt`; Fortran interpolates bilinearly in  $(\log T, \log n_e)$ . Checks:  $4\pi j_{\text{H}\beta}/(n_e n_p) = 1.235 \times 10^{-25}$  at  $(10^4 \text{ K}, 10^2 \text{ cm}^{-3})$  and  $\text{H}\alpha/\text{H}\beta = 2.864$  (canonical values).

He II ( $Z = 2$ ) comes from the *original* CDS file `e2b.d`, whose format differs from the repackaged `e1bx.d` (six-token block headers; index  $k = (n_{\text{cut}} - n_{\text{u}})(n_{\text{cut}} + n_{\text{u}} - 1)/2 + n_{\text{l}}$  with  $n_{\text{cut}} = 25$ , from the SH95 `intrat.f` reader): six lines (1640, 4686, 3204, 10126, 5413, 4543 Å) on a  $12 \times 13$  grid in `data/atomic/sh95_heii_caseB.txt`, weighted in the line table by  $n_e n_{\text{HeIII}}$ . Checks:  $4\pi j_{4686}/(n_e n_{\text{HeIII}}) = 1.495 \times 10^{-24}$  at  $(10^4 \text{ K}, 10^4 \text{ cm}^{-3}) = \alpha_{4686}^{\text{eff}}(3.57 \times 10^{-13}) h\nu$ , and  $1640/4686 = 6.56$  there.



## 4. Algorithms

### 4.1 The iteration loop

```

read grid → persist gas leaf state → fill kap_ion
do iter = 1, gas_niter
  zero the band tally  $J_b$ 
  transport stellar packets (analytic edge walk, Eq. 2)
  [G3] rebuild the diffuse emission table from the current state;
    emit and transport  $L_{\text{dif}}/L_{\text{pkt}}$  diffuse packets
  ALLREDUCE  $J_b$ ; rate integrals (Eq. 3); metal  $\Gamma$ 's
  solve each leaf: ionization only (G1) or coupled ( $x, T_e$ ) (G2)
  refill kap_ion from the new state (the stellar tally is
    recomputed from zero each iteration — no reuse)
  converged by par%conv_crit: cell-max or volume-integrated tests (below)
end do
outputs: rates + state maps; line luminosities; nebular continuum

```

Every rank runs the identical cell solves (inputs are the reduced tallies), so all ranks agree on convergence; only the node master writes the shared state arrays. Under-relaxation on  $x$  (par%ion\_relax) is available for sharp I-fronts. Practical lessons recorded: a fully neutral start advances the front only  $\sim$ one cell per iteration (the photons die in the first neutral cell), while an ionized start converges globally; and  $\max |\Delta x_{\text{HII}}|$  stalls at the front-cell Monte-Carlo noise floor ( $\sim 2\text{--}4 \times 10^{-2}$  at  $4 \times 10^7$  packets), so tolerances below that are unreachable by that criterion — the G3 diffuse field, interestingly, smooths the front and converges to  $3 \times 10^{-3}$ .

**Convergence measures.** Two pairs are computed every iteration and par%conv\_crit selects which gates the loop: the cell-max tests ( $\max |\Delta x_{\text{HII}}| < \text{gas\_tol}$ ,  $\max |\Delta T_e|/T_e < \text{gas\_tol\_te}$ ) and the volume-integrated tests

$$\frac{\sum_{\text{leaf}} V |\Delta x_{\text{HII}}|}{\sum_{\text{leaf}} V x_{\text{HII}}} < \text{gas\_tol}, \quad \frac{\sum_{\text{leaf}} V n_e |\Delta T_e|}{\sum_{\text{leaf}} V n_e T_e} < \text{gas\_tol\_te}. \quad (12)$$

The volume measures fix both stall modes of the cell-max tests: front jitter occupies little volume (the front shell is a few thousand cells against  $\sim 10^5$  ionized ones, so its noise enters the  $L_1$  norm suppressed by that ratio), and the  $n_e$  weight removes the vacuum and no-heating cells whose  $t_{\text{e\_min}}$ -pinned oscillation dominates  $\max |\Delta T_e|$  in FUV/PDR configurations. 'vol' is the recommended production setting; 'cell' (the historical behavior) remains the default so the recorded gates reproduce. Measured on the FUV dusty Strömgren case at  $8 \times 10^6$  packets (where the cell-max tests sat at 0.1 and 6.4 indefinitely): the volume measures decay smoothly to well-defined Monte Carlo floors of  $2.7 \times 10^{-3}$  ( $x$ ) and  $6.6 \times 10^{-3}$  ( $T_e$ ) by iteration  $\sim 27$ , and running 30 further iterations leaves the solution unchanged ( $V_{\text{ion}}$  to 0.014%) — the floor IS convergence. Set the tolerances just above the floor for the packet count in use (here  $5 \times 10^{-3}$  and  $10^{-2}$  would stop the loop at iteration  $\sim 23$ ); the floors scale as  $1/\sqrt{N_{\text{packets}}}$ .

## 4.2 Ionization equilibrium

Per leaf at temperature  $T$  (`solve_ion_cell`): with  $R_i = \Gamma_i + n_e C_i(T)$  the ionization rates and  $\alpha_i$  the case-A/B recombination rates,  $x_{\text{HI}} = \alpha n_e / (R_{\text{HI}} + \alpha n_e)$  and the He stage chain  $n_{i+1}/n_i = R_i / (\alpha_{i+1} n_e)$ , closed by  $n_e = n_{\text{H}} [x_{\text{HII}} + y_{\text{He}} (x_{\text{HeII}} + 2x_{\text{HeIII}})]$  and solved by a damped fixed point on  $n_e$  (mixing  $\frac{1}{2}$ , converged at  $10^{-10}$ ,  $\leq 200$  iterations).

## 4.3 Thermal balance

G2 solves  $\mathcal{H}(T) = \mathcal{C}(T)$  by bisection on  $\log T_e$  in  $[\text{te\_min}, \text{te\_max}]$  ( $\sim 40$  steps to  $10^{-5}$  relative), re-solving the ionization state at every trial  $T$ . Heating is  $n_{\text{HI}} \mathcal{H}_{\text{HI}} + y_{\text{He}} n_{\text{H}} (x_{\text{HeI}} \mathcal{H}_{\text{HeI}} + x_{\text{HeII}} \mathcal{H}_{\text{HeII}})$  from Eq. (3); cooling assembles recombination (Hui–Gnedin  $\beta$ ), free-free ( $1.42554 \times 10^{-27} \sqrt{T} Z^2 \bar{g} n_e n_{\text{ion}}$ ), collisional ionization ( $k_{\text{ci}} \times$  ionization potential), H I line cooling (Tier-1 fit), and the trace-metal sum  $\sum n_e n_{\text{ion}} \Lambda_{\text{ion}}(T)$  with the cascade fractions re-evaluated at each trial  $T$ .

## 4.4 The trace-metal cascade

The registry (`species_mod`) activates an element when its abundance is positive. In the trace approximation (no feedback on  $n_e$  or opacity), stage fractions follow the product chain

$$\frac{n_{i+1}}{n_i} = \frac{\Gamma_i + n_e C_i + n_{\text{HII}} k_{\text{ion},i}^{\text{CX}}}{n_e [\alpha_{\text{RR}} + \alpha_{\text{DR}}]_{i+1} + n_{\text{HI}} k_{\text{rec},i+1}^{\text{CX}}}, \quad (13)$$

with metal  $\Gamma_i$  computed once per iteration from the same band tally. Adding an ion is one entry in the data generators plus its Tier-1/Tier-2 files.

## 4.5 Explicit diffuse field (G3)

Case B on the spot is replaced by case A rates plus explicit emission of ground-recombination packets: three channels with rates  $\alpha_1 = \alpha_A - \alpha_B$  and photon energies  $E = E_{\text{th}} + kT_e x$ ,  $x \sim \text{Exp}(1)$  (the near-threshold Milne shape at bin resolution). Each iteration the emission table  $L_{\text{ch}}(\text{leaf}) = n_e n_{\text{rec}} \alpha_1(T) (E_{\text{th}} + kT) V$  is rebuilt from the current state; packets carry the stellar  $L_{\text{pkt}}$ , sample the leaf from a CDF, a uniform position within it, and are transported by the same edge walk. At HII40 the diffuse band luminosity converges to  $6.8 \times 10^{38} \text{ erg/s} = 0.50 L_*$ .

With `par%hei_diffuse` a fourth channel adds the He I *excited-level* recombination radiation (previously those photons simply vanished): all case-B He II recombinations cascade to  $n = 2$  and decay with the AGN3 low-density branching —  $3/4$  to  $2^3\text{S}$  (a single 19.82-eV photon),  $1/4 \times 2/3$  to  $2^1\text{P}$  ( $584 \text{ \AA} = 21.22 \text{ eV}$ ),  $1/4 \times 1/3$  to  $2^1\text{S}$  (two-photon; probability 0.56 of an H-ionizing photon, sampled flat in  $[13.6, 20.62] \text{ eV}$ ). At HII40 this raises the diffuse luminosity by 4.7% and  $T_e$  by 1.1% while leaving the He I/II split unchanged to four decimals — correct energy bookkeeping, and the demonstration that the He-structure residual against MOCASSIN does NOT come from this channel (the split is set by the  $>24.6\text{-eV}$  budget).

## 4.6 Solution-driven I-front re-refinement (G4)

At iteration `refine_iter` the octree is rebuilt from the current solution: an old leaf is flagged as front when  $\epsilon < x_{\text{HI}} < 1 - \epsilon$  or a face neighbor differs by more than `refine_dx`; the new tree carries level `refine_lmax` wherever a cell overlaps

a front-flagged leaf and `refine_lbase` elsewhere. The gas state and `rhokap` map over by position (each new leaf samples the old leaf containing its center), and every dependent array is recreated (the band opacity, tallies, metal  $\Gamma$  blocks, diffuse tables, rate arrays — the last was the one initially missed, an allocate-once array that faulted when  $n_{\text{leaf}}$  grew). All ranks rebuild identically from the shared state, and the old grid’s shared-memory windows are *recycled*: `destroy_shared_mem` (the collective counterpart of `create_shared_mem`) frees the 18 old windows right after the position sampling — the old tree’s last reader — so repeated re-refinement neither leaks nor holds old and new grids in memory simultaneously.

## 4.7 Dust in the ionizing band

The dust coupling of PLAN section 7 items 1–3, now complete:

- **Absorption competition.** With `par%ion_add_dust`, each band bin adds  $\kappa_d(b, \text{leaf}) = \rho \kappa_{\text{leaf}} s_{\text{abs}}(b)$  where  $s_{\text{abs}}(b) = (1 - a(E_b)) C_{\text{ext}}(E_b) / C_{\text{ext}}(\lambda_{\text{ref}})$  comes from a `kext` table covering the EUV (the WD01/D03  $R_V=3.1$  table,  $1 \text{ \AA} - 10^4 \mu\text{m}$ :  $C_{\text{abs}}/\text{H} = 1.83 \times 10^{-21}$  at 13.6 eV falling to  $3.96 \times 10^{-22} \text{ cm}^2$  at 100 eV). A trap worth recording: the D03 table is *not monotonic* in wavelength (an out-of-order seam) — without sorting, the linear interpolation search silently returned  $s_{\text{abs}}(100 \text{ eV}) = 0.903$  instead of 0.832 and shifted the dusty Strömgren radius by +1.6%.
- **Scattering.** With `par%ion_dust_scatter` the scattering coefficient joins the extinction ( $s_{\text{sca}}(b) = a C_{\text{ext}} / C_{\text{ext,ref}}$ ; in-band Henyey–Greenstein asymmetry  $g(E_b) = 0.67 - 0.99$  from the table  $\langle \cos \rangle$  column) and interactions are sampled: after the analytic zero-variance direct walk, a first interaction is forced along the ray with weight  $w_1 = 1 - e^{-\tau_{\text{edge}}}$ , the packet keeps the albedo fraction  $a_{\text{eff}} = \kappa_{\text{sca,d}} / \kappa_{\text{ext}}$  at each interaction, scatters into an HG direction, and continues with unforced flights (whose segments tally into  $J_b$ ) under Russian roulette ( $w_{\text{min}} = 10^{-4}$ ). Stellar and diffuse packets share the walk; gas ionization, gas heating, and grain heating all derive from the same tally, so the energy split is bookkept by construction.
- **Live ionization-dependent dust with a PAH split.** `dust_model='laursen09_live'` recomputes the dust density from the *computed*  $x_{\text{HI}}$  every iteration,  $n_d \propto (Z/Z_{\text{ref}}) n_{\text{H}} [(1 - f_{\text{PAH}})(x_{\text{HI}} + f_{\text{ion},x_{\text{HII}}}) + f_{\text{PAH}}(x_{\text{HI}} + f_{\text{ion,PAH},x_{\text{HII}}})]$ , the PAH share carrying its own ionized-gas survival ( $f_{\text{ion,PAH}} = 0$  default: PAHs destroyed in ionized gas) — the PLAN 7-3 generalization including its PAH-specific option.
- **Grain heating and the equilibrium temperature.**  $\text{Heat\_dust}(\text{leaf}) = \sum_b [4\pi J \Delta\nu]_b \kappa_{\text{abs,d}} [\text{erg s}^{-1} \text{ cm}^{-3}]$  is consumed by `dust_temp_mod`:  $f_{\text{emit}}(T) = 4\pi \int s_{\text{abs}}(\nu) B_{\nu}(T) d\nu$  is precomputed from the same `kext` table (log- $T$  grid, 3.2–3200 K) and  $T_d(\text{leaf})$  solves  $\text{Heat\_dust} = (\rho \kappa / d_{\text{cm}}) f_{\text{emit}}(T_d)$ ; outputs are the  $T_d$  leaf map and the grid-integrated modified-blackbody IR spectrum ( $1 - 1000 \mu\text{m}$ ). FUV grain heating is transported by the `par%add_fuv` band extension (Section 2.1): gas cross sections vanish below their thresholds automatically, so no new transport code is involved, and the two-segment grid keeps the ionizing bins at full resolution. This is the single equilibrium *mixture* temperature; the size- and material-resolved treatment is the SEDust coupling below.

## 4.8 Stochastic dust emission (SEDust coupling)

With `par%dust_sed`, `sedust_mod` (adapted from the MoCafe `dustemis_mod`) computes the grid-integrated dust SED with size-resolved stochastic heating and PAH bands, using the SEDust library (`Libsedust.a` + module files copied to `SEDust_lib/`; `astrodust`, `DL07`, and `Zubko` grain models). The coupling follows the MoCafe convention that makes the normalization exact by construction:

- **Shape from the local field.** For each leaf, the band tally  $J_b$  is converted to  $J_\lambda$  on the SEDust wavelength grid (1129 points,  $0.0912\text{--}4 \times 10^4 \mu\text{m}$  for `astrodust`). The band reaches into the EUV, below the SEDust optics grid; that energy is deposited into the shortest SEDust bin, conserving the absorbed power (the hardness of the stochastic temperature spikes is slightly underestimated there — a documented approximation). SEDust then returns the emission spectrum of the grain population in this field: equilibrium large grains plus the full temperature-distribution solve for small grains and PAHs.
- **Normalization from the transport.** The SEDust spectrum is used only as a shape: each leaf's emission is normalized to its absorbed power  $\text{Heat\_dust} \cdot V$  from the Monte Carlo tally (radiative equilibrium). H II regions are optically thin to the re-emitted IR, so no Lucy re-iteration is needed and the grid-integrated  $L_\lambda$  is accumulated directly (an `MPI_ALLREDUCE` over leaves distributed round-robin).  $\int L_\lambda d\lambda = \sum \text{Heat\_dust} V$  holds by construction and is written into the `_dustsed.txt` header.
- **Cost.** The exact stochastic solve costs  $\sim 0.1\text{--}0.2$  s for each leaf with nonzero grain heating; it runs once, at output time, after convergence (the dusty Strömgren smoke test:  $\sim 1.9$  hours on 8 ranks for a  $64^3$  grid — the solve, not the transport, dominates).
- **Smoke result.** Dusty Strömgren at  $4 \times 10^4$  K:  $\int L_\lambda d\lambda = 4.296 \times 10^{38} \text{ erg/s} = \sum \text{Heat\_dust} V$  to 4 digits; the SED shows the PAH bands at  $3.3/6.2/7.7/8.6/11.3/12.7/17 \mu\text{m}$  over a warm  $20\text{--}60 \mu\text{m}$  plateau with the FIR falloff — the canonical stochastically heated spectrum for a hot compact field.
- **Live PAH share** (`par%sed_pah_live`): `dust_emission`'s channel output (`astrodust`: 'AD' + 'PAH') lets each leaf's spectrum be assembled with the PAH channel weighted by the leaf's PAH survival ( $x_{\text{HI}} + f_{\text{ion,pah}} x_{\text{HII}}$ ), divided under `laursen09_live` by the dust survival already carried by `rhokap`; the  $\text{Heat\_dust} \cdot V$  normalization keeps energy closure (the PAH energy shifts to the large grains). Smoke: the PAH bands drop to  $0.26$  ( $7.7 \mu\text{m}$ ) and  $0.15$  ( $11.3 \mu\text{m}$ ) of the PAH-everywhere reference, the FIR rises by  $1.06\text{--}1.36\times$ , and the  $8\text{-}\mu\text{m}$  map turns from centrally peaked into a *ring* at the ionization front — the classic Spitzer/JWST bubble morphology.

## 4.9 Peel-off imaging

With `par%ion_peel`, one extra transport pass runs on the converged state with observer peeling (the MoCafe peeling-off convention, adapted to the band): every emitted packet — stellar and diffuse — peels a direct contribution  $L_{\text{pkt}} w e^{-\tau} / (4\pi r^2)$  toward each observer, and, when dust scattering is sampled, every interaction peels a scattered contribution with the Henyey–Greenstein phase function of the packet's bin (after the albedo weight, before the new direction is sampled). The optical depth to the box edge along the observer direction is integrated through `kap_ion` at

the packet's bin by a tally-free walk (tallying the virtual peel ray would double count), so the images carry exactly the transport's extinction: gas, dust absorption, and dust scattering. Images are TAN projections on the MoCafe observer geometry (obsx/y/z or angle lists, rotation matrix, auto-sized pixel scale), written as surface brightness [ $\text{erg s}^{-1} \text{cm}^{-2} \text{sr}^{-1}$ ] split into direct/scattered and ionizing/FUV channels. The band tally is rebuilt by the imaging pass, so the written rates reflect the same field the images carry. With `par%peel_bins` the channel axis becomes the band bin — `direct_cube/scatt_cube` ( $n_x, n_y, n_\nu$ ) — for hardness maps and synthetic EUV/FUV photometry (the band-integrated channel images are still written, derived from the cube; the gate shows the expected color: the FUV scattered halo shrinks from a 15.5-px to a 12.0-px half-light radius between 6.2 and 13.2 eV).

For the *dust* bands no transport is needed — the nebula is optically thin to its own IR — so `par%dust_emis_bands` instead stores SEDust band emissivities of each leaf (`_dustemis`, the same block layout as the `_emis` file) and the Python map maker projects them into flux-conserving dust-band images exactly like line maps.

#### 4.10 Uniform grids with DDA traversal

`par%grid_type = 'uniform'` accepts the same single-level leaf list and stores it in raster order: `leaf = cell = 1 + i_x + n_x(i_y + n_y i_z)`. No tree topology and no neighbor table are built; the leaf finder becomes three integer divisions and the face-neighbor step becomes integer arithmetic — together with the exact plane-crossing step the walks already compute, this IS the Amanatides–Woo DDA traversal. Everything downstream is untouched (the gas modules only consume leaf-indexed arrays), `refine_front` is octree-only, and the memory saved is the tree itself (children  $8\times$ , neighbor  $6\times$ , parent, `ileaf` maps — noticeable at  $512^3$  and above). Gate: the same-seed Strömgen run on 'uniform' reproduces the single-level-octree run *bit-identically* leaf by leaf ( $\max|\Delta x_{\text{HI}}| = 0$ ; the transport is deterministic and the summation order along each ray is unchanged), with a  $\times 1.13$  transport speedup — modest, as expected, since the neighbor-table walk was already  $O(1)$  per cell.

#### 4.11 PDR-zone physics

Three switches (default off) extend the FUV zone beyond ion structure:

- `par%metal_ne`: electrons from the metal cascade join the  $n_e$  closure of the cell solve,  $n_e = n_{\text{H}}[x_{\text{HII}} + y_{\text{He}}(\dots)] + \sum_{\text{el}} A_{\text{el}} n_{\text{H}} \sum_i (i-1) f_i$  — a  $\lesssim 10^{-3}$  correction inside the H II region but the *only* electrons beyond the front ( $n_e \approx n_{\text{CII}} \approx A_{\text{C}} n_{\text{H}}$  in the PDR zone), where they control the [C II]/Mg II excitation and the photoelectric efficiency.
- `par%metal_heat`: metal photoheating enters the thermal balance; the heating integrals  $\mathcal{H}_i = \sum_b [4\pi J \Delta\nu]_b \sigma_i (1 - E_{\text{th},i}/E_b)$  are accumulated together with the metal  $\Gamma$ 's.
- `par%grain_pe`: grain photoelectric heating and grain recombination cooling with the Bakes & Tielens (1994) fits,  $\Gamma_{\text{pe}} = 1.3 \times 10^{-24} \epsilon G_0 n_{\text{H}}$  with  $\epsilon = 4.87 \times 10^{-2} / [1 + 4 \times 10^{-3} x^{0.73}] + 3.65 \times 10^{-2} (T/10^4)^{0.7} / [1 + 2 \times 10^{-4} x]$ ,  $x = G_0 \sqrt{T} / n_e$ , and  $\Lambda_{\text{rec}} = 4.65 \times 10^{-30} T^{0.94} x^\beta n_e n_{\text{H}}$ ,  $\beta = 0.74/T^{0.068}$ , both scaled by `pe_scale` times the leaf's dust amount relative to Milky-Way dust, measured from the leaf opacity itself:  $(\rho\kappa/d_{\text{cm}})/(n_{\text{H}} C_{\text{ext,MW}}(V))$  with  $C_{\text{ext,MW}}(V) = 4.868 \times 10^{-22} \text{cm}^2/\text{H}$  — robust against the DGR-versus-cext\_dust input conventions (the first gate run had DGR = 1 with the MW extinction folded into `cext_dust`; a naive DGR/0.01 scale made the heating

100× too strong and drove the zone to a false Ly $\alpha$ -capped 12 kK equilibrium).  $G_0$  is the local FUV field in Habing units summed from the FUV bins of the tally (add\_fuv required). Three further guards, each found by the gate: (i) the BT94 fits are evaluated at  $\min(T, 10^4 \text{ K})$  — outside their fit range the  $T^{0.7}$  term of  $\epsilon$  diverges; (ii) both grain terms carry an  $x_{\text{HI}}$  weight — the BT94 charging parameter knows only the FUV field, so in ionized gas (EUV-charged grains, grain energy already routed to the IR via Heat\_dust) it would overheat badly; (iii) the package includes the *H-impact* fine-structure cooling of [C II] 158  $\mu\text{m}$  and [O I] 63  $\mu\text{m}$  (two-level, low-density limit;  $q_{ul}^{\text{H}} = 7.6 \times 10^{-10} (T/100)^{0.14}$  and  $9.2 \times 10^{-11} (T/100)^{0.67} \text{ cm}^3 \text{ s}^{-1}$ , Barinovs et al. 2005) — the Tier-1 fits are electron-impact only, and with  $n_{\text{HI}}/n_e \sim 10^3\text{--}10^4$  in the PDR zone the H collisions ARE the coolant. Without (iii) the zone has no coolant between the fine-structure and Ly $\alpha$  regimes, and the  $\epsilon(x)\text{--}n_e$  feedback (collisional ionization raises  $n_e$ , neutralizing the grains and raising  $\epsilon$  fifty-fold) drives it to a false  $\sim 10^4 \text{ K}$  equilibrium.

Scope note: this is the atomic PDR skin — no H<sub>2</sub>/CO chemistry, no line transfer, no dust–gas collisional coupling — so PDR temperatures are first-order estimates, adequate for the ion structure and rough [C II] emissivities.

#### 4.12 Tier-2 $n$ -level solver and line luminosities

nlevel\_mod evaluates exactly the recipe of Section 3.7 (Clenshaw evaluation of the Chebyshev series) and solves the statistical equilibrium  $\dot{n} = 0$ ,  $\sum_i n_i = 1$  (partial-pivot elimination; de-excitation  $q_{ul} = 8.629 \times 10^{-6} \Upsilon / (g_u \sqrt{T})$ , excitation by detailed balance). lines\_mod integrates  $L_{\text{line}} = \sum_{\text{leaf}} j n_{\text{ion}} V$  on the converged state and normalizes to the SH95 H $\beta$ . The Fortran chain was verified against the Python reference solver on an identical state: 26 line luminosities agree to four significant digits. With par%emis\_output the same solve also writes the emissivities leaf by leaf (\_emis: an emis\_<el>\_<stage> block for each ion and emis\_h for the SH95 lines, [ $\text{erg s}^{-1} \text{ cm}^{-3}$ ] with  $\sum \text{emis} \cdot V = L_{\text{line}}$ , verified to  $4 \times 10^{-4} \%$ ), together with n\_H,  $n_e$ ,  $T_e$ , the H/He fractions, and the metal stage fractions — a self-contained input for line maps and position–velocity work.

### 5. Validation gates and results

#### 5.1 G0: rate integrals vs. analytic attenuation

Uniform sphere ( $n_{\text{H}} = 1$ ,  $R = 1 \text{ pc}$ ,  $x_{\text{HI}} = 0.1$ ,  $y_{\text{He}} = 0.1$ ), central  $10^5 \text{ K}$  source, 16 bins,  $10^8$  packets. Against the cell-volume-averaged analytic attenuation with identical bins:  $\Gamma_{\text{HI}}$  mean bias +0.004%, median 0.38% (pure Monte Carlo noise), max 2.6%;  $\Gamma_{\text{HeI}}$  bias +0.011% (Figure 1).

#### 5.2 G1: Strömgren sphere

Uniform  $n_{\text{H}} = 100$  sphere,  $Q_{\text{H}} = 10^{49} \text{ s}^{-1}$  ( $T_{\star} = 4 \times 10^4 \text{ K}$ , 32 bins), case B,  $T_e = 10^4 \text{ K}$ . The reference is an exact 1D radial marching with identical bins and rates — necessarily *Gauss–Seidel* (each shell solved with the already-updated interior opacity): a Jacobi sweep advances the front only one absorption length per pass and never converges. Sharp-edge analytic  $R_{\text{S}} = 3.153 \text{ pc}$ ; the 1D reference gives  $R_{\text{eff}} = (3V_{\text{ion}}/4\pi)^{1/3} = 3.0365 \text{ pc}$  (the  $\sim 4\%$  gap is physical: He consumes photons and the model re-emits none).

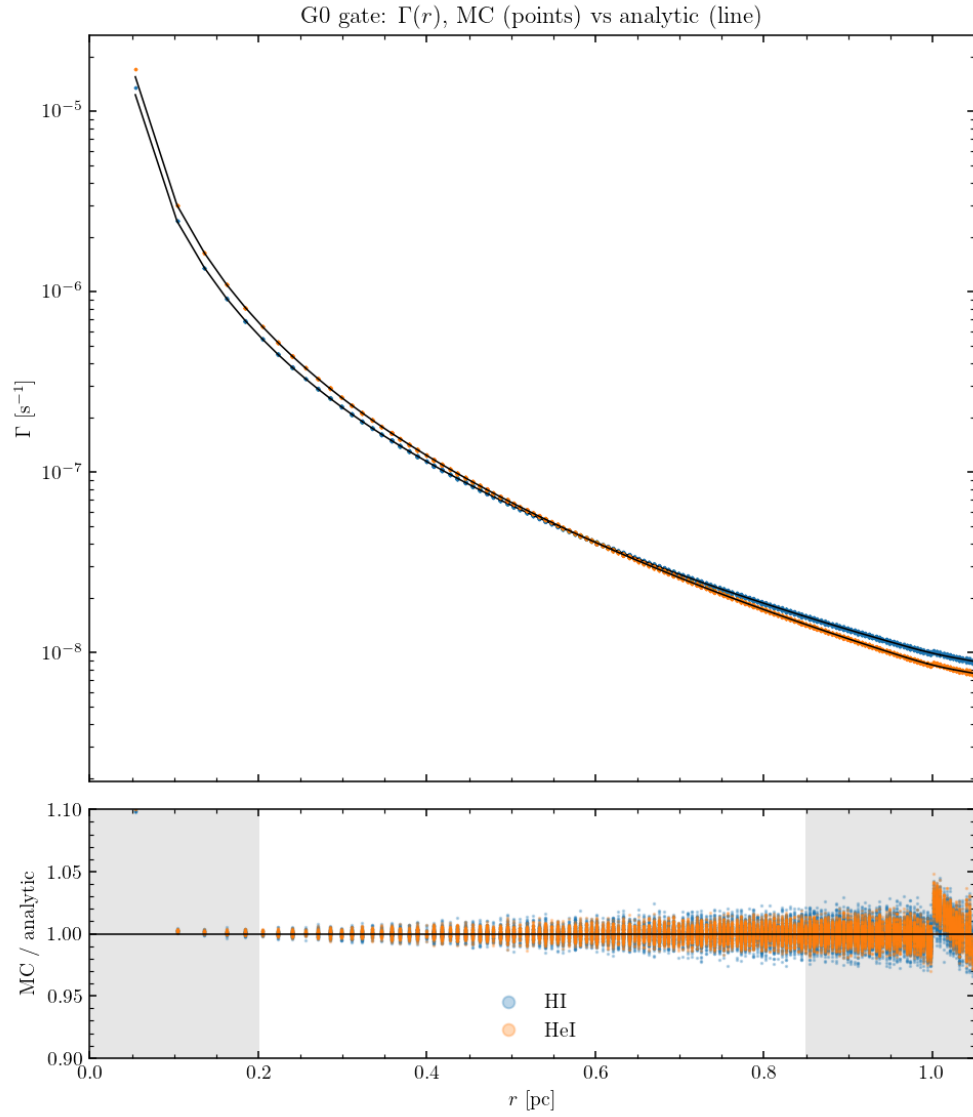


Figure 1: G0 gate:  $\Gamma(r)$  from the Monte Carlo tally (points) against the analytic point-source attenuation (lines); the lower panel shows the ratio (shaded regions excluded: cell-size effects at the center, the voxelized sphere edge outside).

| run                       | leaves  | $R_{\text{eff}}$ [pc] | vs. 1D |
|---------------------------|---------|-----------------------|--------|
| uniform level 6           | 262 144 | 3.0534                | +0.56% |
| front-refined levels 4–6  | 74 320  | 3.0534                | +0.56% |
| center-refined levels 4–6 | 7 568   | 3.1078                | +2.35% |

The front-refined octree reproduces the uniform grid to four decimals with  $3.5\times$  fewer cells; the center-refined control (coarsest cells *at* the front) shows why the I-front is the one place needing resolution (Figure 2).

### 5.3 G2a: thermal H/He nebula

Same configuration with the thermal balance on (Figure 3):  $T_e(r)$  matches the 1D thermal reference to median 0.136%, mean bias +0.006%, max 0.95% over  $0.3 < r < 2.8$  pc — including the central He III-zone temperature structure and the front drop. Without metals the nebula runs at 19–24 kK, the known pure-H/He overshoot that motivates the G2 metal frame.

### 5.4 G2c/G3: the MOCASSIN Lexington benchmarks

HII20/HII40: blackbody 20/40 kK,  $Q_{\text{H}} = 1.006/4.26 \times 10^{49} \text{ s}^{-1}$ ,  $n_{\text{H}} = 100$ , hollow shell  $R_{\text{in}} = 3 \times 10^{18}$  cm, abundances He 0.1, C  $2.2 \times 10^{-4}$ , N  $4 \times 10^{-5}$ , O  $3.3 \times 10^{-4}$ , Ne  $5 \times 10^{-5}$ , S  $9 \times 10^{-6}$ . References are the stored MOCASSIN 2.02.73.2 baselines (reduced runs:  $10^5$  photons, 3 iterations, convergence flag 0.09).

**Temperatures.**  $T_e$  weighted by  $n_{\text{HII}}n_e$ :

|       | MoCHII case-B OTS | MoCHII case A + diffuse | MOCASSIN |
|-------|-------------------|-------------------------|----------|
| HII20 | 7257 K (+0.11%)   | 7372 K                  | 7249 K   |
| HII40 | 8991 K (+7.0%)    | 9130 K                  | 8404 K   |

The two MoCHII treatments bracket a converged answer near 9.0–9.1 kK for HII40: the transport treatment moves  $T_e$  by only  $\sim 1.5\%$  and is *not* the residual’s origin. The audited cooling budget at mid-nebula ([O III] 40%, [S III] 13%, [O II] 10%, recombination 12%, free-free 10%) is sane; the attribution is atomic-data vintage — MOCASSIN’s `oiii.dat` header states CHIANTI *version 7*, and a 20% collision-strength shift on the dominant coolant moves  $T_e$  by  $\Delta T \sim \Delta \ln \Upsilon T^2 / (E/k) \approx 550$  K — combined with the underconverged baseline. (Published Lexington-2000 HII40 spread:  $\sim 8300$ – $8700$  K.)

**Ionization structure** ( $n_e$ -weighted means, HII40, after the KF96 charge exchange of Section 3.5 was added):

| ion   | MoCHII | MOCASSIN | ion    | MoCHII | MOCASSIN |
|-------|--------|----------|--------|--------|----------|
| H I   | 0.0032 | 0.0035   | O II   | 0.371  | 0.512    |
| He II | 0.991  | 0.941    | O III  | 0.627  | 0.484    |
| C II  | 0.242  | 0.266    | Ne II  | 0.647  | 0.659    |
| C III | 0.751  | 0.728    | Ne III | 0.352  | 0.334    |
| C IV  | 0.0057 | 0.0054   | S II   | 0.243  | 0.061    |
| N II  | 0.228  | 0.344    | S III  | 0.700  | 0.794    |
| N III | 0.771  | 0.649    | S IV   | 0.057  | 0.142    |



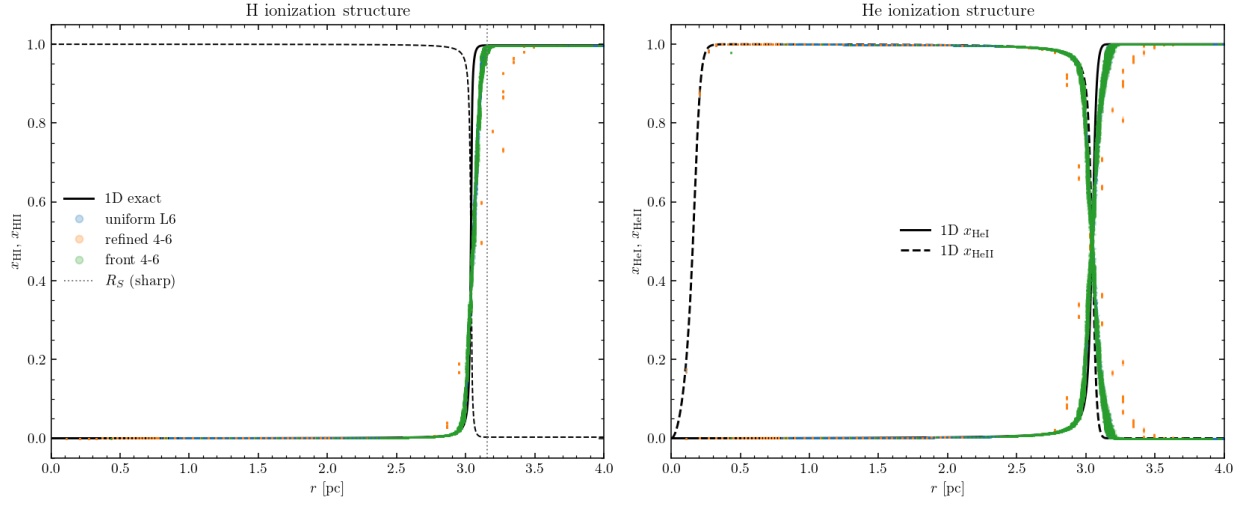


Figure 2: G1 gate: H and He ionization structure against the 1D exact reference (black); uniform level-6 (blue), center-refined (orange), front-refined (green, overlying blue).

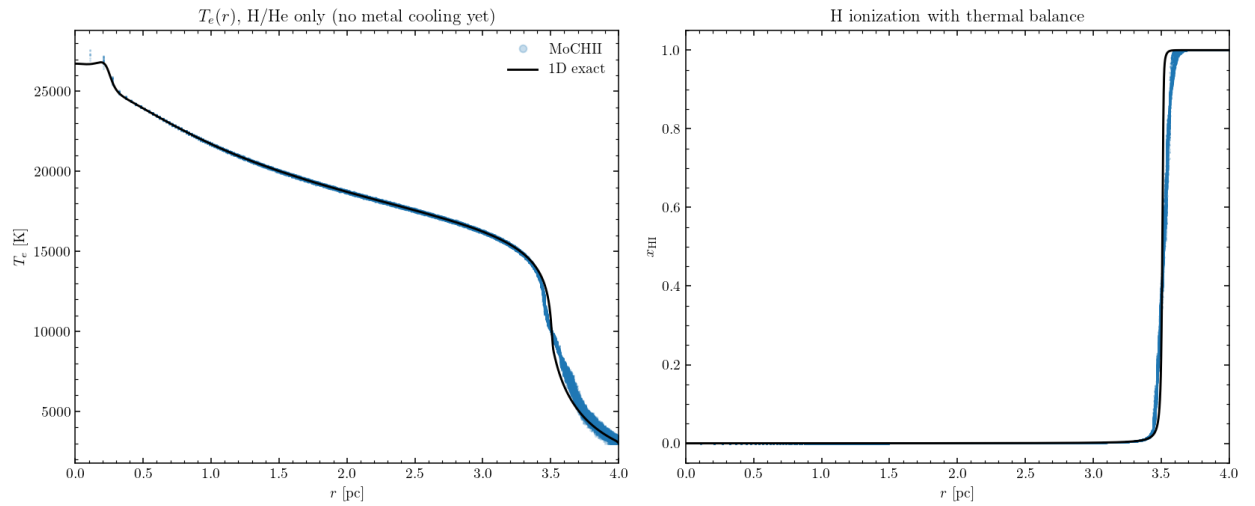


Figure 3: G2a gate:  $T_e(r)$  and  $x_{\text{HI}}(r)$  with the thermal balance, against the 1D exact reference.

The KF96 addition was decisive (before: O III 0.79, Ne II 0.44, C III 0.83). At HII20 the structure matches everywhere except Ne I/II (0.04/0.96 vs. 0.46/0.54; the 21.6-eV threshold at a 20-kK source is a known code-spread hotspot) and the S II/III split.

**Line fluxes** (ratios to  $H\beta$ ; MoCHII from the G3 runs): absolute  $L(H\beta)$  agrees to  $-0.7\%$  (HII40) and  $-4.6\%$  (HII20). At HII40, [Ne III] 15.5  $\mu\text{m}$  matches to 0.3% and [Ne II] 12.8  $\mu\text{m}$  to 9%; every discrepant line is quantitatively the structure/ $T_e$  difference above, e.g. [O III] 5007 at ratio  $1.79 = 1.27$  (fraction excess)  $\times 1.32$  ( $e^{-E/kT}$  at 9130 vs. 8404 K), and [S II] 6716 at  $3.6 =$  the S II fraction ratio. The emissivity chain itself is exact (Section 4.12).

## 5.5 G4: re-refinement gate and the TNG demonstration

The re-refinement of Section 4.6 on the G1 Strömgren configuration (level-5 start, rebuild to level 7 at iteration 15, 32 768  $\rightarrow$  464 248 leaves):

| run                          | leaves    | $R_{\text{eff}}$ [pc] | vs. 1D |
|------------------------------|-----------|-----------------------|--------|
| re-refined 5 $\rightarrow$ 7 | 464 248   | 3.0432                | +0.22% |
| native uniform level 7       | 2 097 152 | 3.0430                | +0.21% |

The two agree to  $+0.008\%$  with  $4.5\times$  fewer leaves, and both improve on level 6 ( $+0.56\%$ ) — resolution-convergent, as the G1 gate wording requires. The front profiles overlies exactly.

The post-processing demonstration runs MoCHII on an Illustris-TNG 60-kpc cutout (the MoCafe converter output,  $n_{\text{H}}$  and  $x_{\text{HI}}$  columns from the simulation) with a  $Q_{\text{H}} = 10^{53} \text{ s}^{-1}$ , 40-kK source at the density peak, thermal balance + metals + re-refinement (Figure 4): dense clumps cast sharp neutral shadows, the circumgalactic gas sits at  $x_{\text{HI}} \sim 10^{-4}$  with  $n_e$ -weighted mean  $T_e = 8916 \text{ K}$ , and the refinement (32 768  $\rightarrow$  599 614 leaves) tracks the *distributed* ionization topology of a turbulent medium — clump skins and partially ionized gas, not a single thin shell. The run also demonstrates why the max-cell convergence test stalls in such media (clump-surface jitter); a volume-integrated criterion is queued.

## 5.6 Dusty Strömgren gate

With the MW D03 dust of Section 4.7 added to the G1 configuration ( $\tau_{\text{d,abs}}(13.6 \text{ eV}) = 1.7$  over the dust-free radius), against the 1D reference carrying the identical dust absorption in every bin:

| case           | $R_{\text{eff}}(1\text{D})$ [pc] | $R_{\text{eff}}(\text{MC})$ [pc] | dev    |
|----------------|----------------------------------|----------------------------------|--------|
| dust-free (G1) | 3.0365                           | 3.0534                           | +0.56% |
| full MW dust   | 2.2307                           | 2.2461                           | +0.69% |
| laursen09_live | 3.0229                           | 3.0397                           | +0.55% |

The classic dusty reduction ( $R/R_0 = 0.735$ ) is reproduced within the gate tolerance; the live ionization-dependent dust drives itself to near dust-free ( $R/R_0 = 0.996$ : interior dust vanishes with the computed  $x_{\text{HII}}$ ), confirming the dust–ionization feedback loop converges self-consistently. Energy budget: dust absorbs 59% of the band luminosity in the full-dust case.

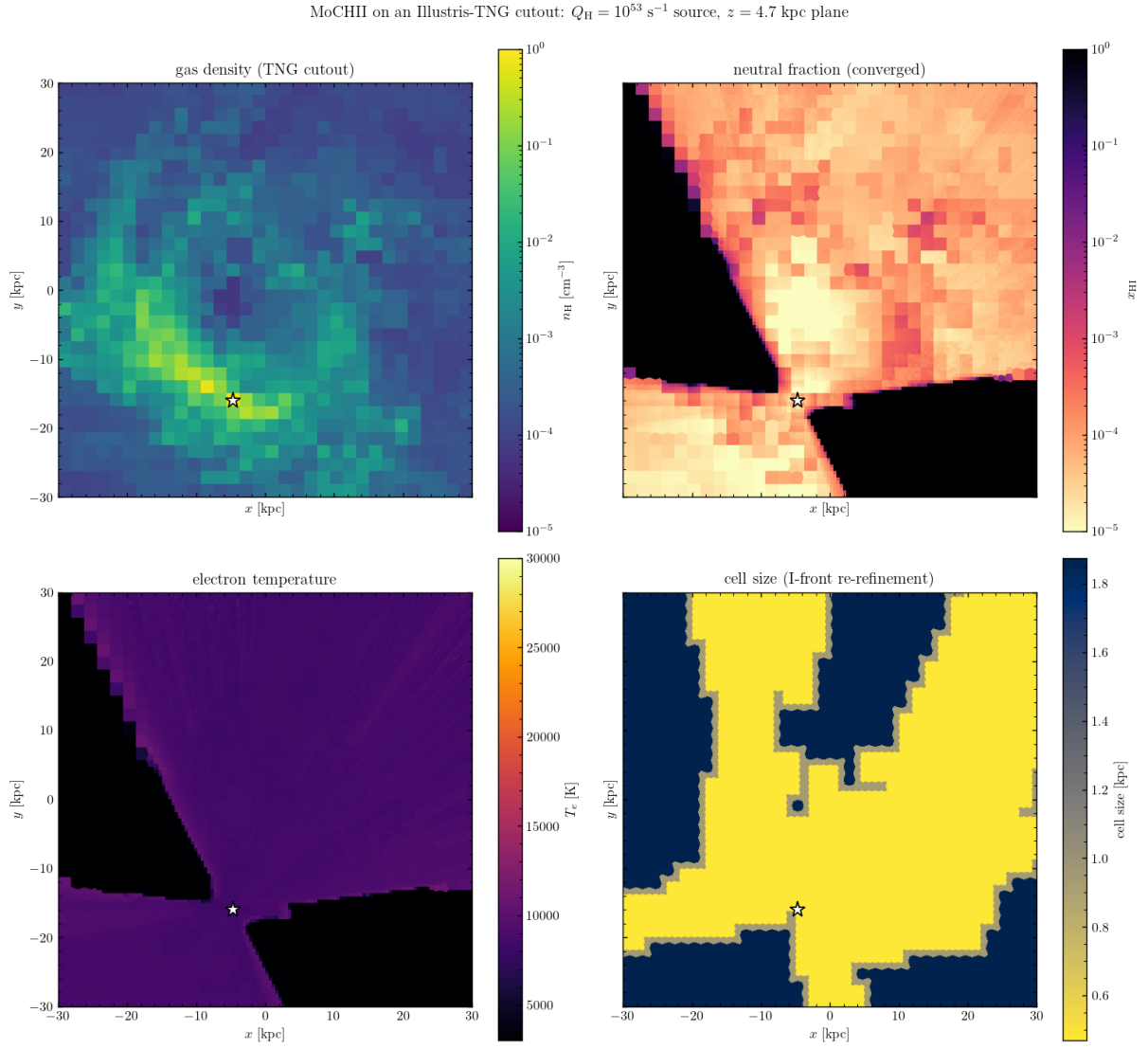


Figure 4: MoCHII on an Illustris-TNG cutout: gas density, converged neutral fraction (note the clump shadows), electron temperature, and the re-refined cell size, in the source plane.

**Scattering bracket.** Turning the scattering on must land the sphere between two analytically defined bounds: treating the dust extinction entirely as absorption (scattered photons destroyed; 1D  $R_{\text{eff}} = 2.086$  pc) and the absorption-only treatment (scattered photons never interact; 2.231 pc, MC counterpart 2.246 pc). The full-scattering run gives  $R_{\text{eff}} = 2.237$  pc — inside the bracket and close to the absorption-only bound, exactly as the strong forward scattering ( $g = 0.67\text{--}0.99$ ) predicts: forward-scattered photons continue nearly radially, and only the small diffused fraction is lost from the radial budget. The dust-absorbed luminosity rises by 0.7% (rescattered photons reabsorbed) — energy bookkeeping consistent.

**Dust temperature and IR.** With the band widened to 6 eV (the band luminosity rescaled by the exact blackbody ratio 2.0969 so  $Q_{\text{H}}$  is preserved) and scattering on, the equilibrium mixture temperature runs from  $T_{\text{d}} = 89$  K at 0.3 pc to 28 K at the sphere edge — the canonical HII-region gradient — and the IR spectrum integral closes on the absorbed power to  $L_{\text{IR}}/L_{\text{abs}} = 1.000$ . (This demo predates the two-segment add\_fuv grid and rescaled the single-segment band by hand; the +1.5% bin-coarsening shift it measured is what the two-segment grid now removes.)

## 5.7 FUV extension gate: PDR-side C II and Mg II

The add\_fuv smoke test (dusty Strömgren, 32 ionizing + 16 FUV bins, thermal + metals with Mg/Fe, dust scattering on; tests/d\_dusty/d\_fuv\_opt.in) verifies the band mechanics and the physics it was built for. Mechanics: the two-segment grid reports  $L_{\text{FUV}}/L_{\text{ion}} = 1.0969$  for the  $4 \times 10^4$  K Planck source at a 6 eV edge — exactly the blackbody band ratio applied by hand in the earlier demo — with  $Q_{\text{H}}$  preserved by construction. Physics: inside the front (2.2 pc with this dust and cooling) the metals sit at C III/Mg III as before; *beyond* it, where H is neutral and the ionizing bins are exhausted, the FUV bins propagate under dust extinction alone and hold the low-threshold stages fully singly ionized — radially averaged C II = 1.000 and Mg II = 1.000 from the front out to the sphere edge, the PDR-side structure that the ionizing-only band cannot produce (there the same zone recombines to C I/Mg I). Two documented limitations in that zone: electrons from metal ionization are not fed back into  $n_e$  (the trace-species design), and with photoheating  $\approx 0$  the thermal bisection pins  $T_e$  at  $\text{te\_min}$  — PDR temperatures require photoelectric heating, outside the present scope. The front-zone cells with vanishing heating also dominate the  $\max |\Delta T_e|$  convergence test (another argument for the queued volume-integrated criterion); the H II-interior structure and  $T_e$  (8.1–10.5 kK) converge normally.

## 5.8 G5: argon as a data operation

Adding argon touched no algorithm: two Tier-1 fits (Ar III, Ar IV), two Tier-2 files, one element file (VFKY96 rows Z=18, Voronov, Badnell RR + SVS82 DR2, KF96 CX), one abundance parameter. PyNeb cross-check: [Ar III] 7136/7751 within 0.7%; [Ar IV] 4740/4711 within 4.0% (a density pair — the same database-spread class as [O II] and [S II]; PyNeb uses RB97/RGJ19 there). End-to-end smoke test: [Ar III] 8.99  $\mu\text{m}$ , 7136, 7751 appear in the line table with 7136/7751 = 4.17, the A-ratio, as expected.

## 5.9 G5 continued: magnesium and iron

Mg and Fe repeat the recipe with three new data wrinkles, all absorbed by the readers: the Fe II power-law RR (RR2), the SVS82-form DR (DR2), and the Mg/Fe charge-exchange forms and stage attribution (Section 3.5). Products: Tier-1 fits Mg II (0.05% max error), Fe II (0.11%), Fe III (0.42%); Tier-2 files Mg II (3 levels), Fe II (13 levels, the IR/optical

forbidden lines), Fe III (14 levels); element files with VFY96  $Z=12$ , 26 rows. PyNeb cross-check: the [Fe III] 4658/4702 ratio agrees to  $-6\%$  to  $-16\%$  — the largest gate deviation so far, and a known property of iron: PyNeb’s default Fe III collision strengths (BB14) differ from CHIANTI v11’s by tens of percent (the iron collision-strength spread in the literature), compounded by the 14-level truncation of a 34-level model. The number is recorded as a data-vintage bracket, not a code error — the same chain reproduces [O III] to 0.6%. End-to-end smoke test (HII20 sphere with  $n_{\text{Mg}}/n_{\text{H}} = 3 \times 10^{-6}$ ,  $n_{\text{Fe}}/n_{\text{H}} = 3 \times 10^{-7}$ , depleted gas-phase values): Mg loads 3 stages, Fe 4 stages; the line table gains the Mg II 2796/2803 doublet (ratio 1.98, the A-ratio limit) and the [Fe II] 1.257  $\mu\text{m}$ , 5.34  $\mu\text{m}$ , and 26  $\mu\text{m}$  lines.

### 5.10 G5 continued: silicon, chlorine, calcium (five stages)

Si, Cl, and Ca bring the registry to eleven elements, each tracked through five stages. The one framework extension was chlorine’s cross section (the VY95 path of Section 3.1 — Cl is absent from the VFY96 outer-shell table); Si carries the KF96 charge exchange from the MOCASSIN chex rows (recombination direction), Cl and Ca have no transcription available. Tier-2 files (all  $\leq 0.65\%$  worst transition): Si II/III/IV, Cl II/III/IV, Ca II, Ca V; Cl V and Ca III/IV have no CHIANTI v11 level data and are tracked without lines (the Ar II pattern). Tier-1 fits: Si II 1.1%, Si IV 0.2%, Ca II 0.4%; Si III 12% and the Cl ions 8–9% resist the multi-exponential form — accepted, since at  $\sim 3 \times 10^{-7}$  abundances their share of the thermal balance is  $\sim 10^{-4}$  and the diagnostics run through the exact Tier-2 solve. PyNeb gates: [Cl II] 8579/9124 to 1.5%; Si III] 1892/1883 to 6.9% and [Cl III] 5518/5538 to 6.6% (PyNeb’s DK94/BZ89 compilations against CHIANTI v11 — the established database-spread class). End-to-end smoke: [Si II] 34.8  $\mu\text{m}$  enters the line table at 4.4% of  $\text{H}\beta$  (the strongest metal line after [C II] — the ionization-front/PDR coolant it is), with Si III] 1883/1892, Si IV 1394/1403, [Cl II] 8581/9126, and the [Cl III] pair at 5518/5538 = 1.402, the low-density limit.

### 5.11 Line and continuum outputs

The converged-state outputs now include: the SH95 H I recombination lines (smoke-state check:  $\text{H}\alpha/\text{H}\beta = 2.94$  at  $\sim 7$  kK); the Tier-2 collisional lines for the 23 registry ions (including [C II] 158  $\mu\text{m}$ , C III] 1909, [N III] 57  $\mu\text{m}$ , Mg II 2796/2803, and the [Fe II] IR lines); and the grid-integrated nebular continuum  $L_{\nu}$  (free-bound + free-free from the verified tables and the Milne extension above the edges, plus the three two-photon continua), written on a 400-point  $\log-\nu$  grid from 11.4  $\mu\text{m}$  to 217  $\text{\AA}$ .

## 6. Known limitations and next steps

- He recombination emits only the three ground continua; the He I excited-channel ionizing photons ( $\sim 60\%$  of He recombinations) are not yet re-emitted — relevant to the He I/II split against MOCASSIN.
- The EUV part of the leaf field, below the SEDust optics grid, is folded into the shortest SEDust bin (energy-conserving; spike hardness slightly underestimated). Peel-off images are band-integrated per channel (ionizing/FUV); bin-resolved image cubes are a straightforward extension if needed.
- The Tier-1 fits are low-density-limit; density suppression above  $n_{\text{crit}}$  enters only through Tier-2 at output time. Ar II carries no cooling or lines (no CHIANTI v11 level data).

- PDR zones: with the Section 4.11 switches off (the default), the FUV extension gives the PDR-side *ion structure* only; with them on,  $n_e$ , metal photoheating, and photoelectric heating make the PDR temperatures first-order estimates —  $\text{H}_2/\text{CO}$  chemistry and line-transfer cooling remain out of scope.
- The cell-max convergence test stalls at the front-cell Monte Carlo noise floor (and at clump-surface jitter in turbulent media); the volume-integrated criterion (`conv_crit = 'vol'`) resolves this and is the recommended production setting.
- MOCASSIN references are the reduced baselines; a full-quality HII40 rerun ( $10^6$  photons, 10 iterations) and a CHIANTI v7-vs-v11 O III  $\Upsilon$  comparison would pin the vintage attribution numerically.
- He II recombination lines (SH95  $Z=2$  data are on disk), the van Hoof et al. (2014) Gaunt swap, the Ercolano & Storey (2006) gamma-table swap, and shared-window recycling for repeated re-refinement are small follow-ups.
- Next: SEDust leaf spectra as emission sources for dust-band imaging, an  $x_{\text{HII}}$ -weighted PAH abundance handed to SEDust (the transport-side split of Section 4.7 does not yet reduce the PAH share of the *emission* model), and further G5 ions as science needs them.